

Correlation Illusion Cards Classroom Run Kit

TOK Cognitive Science Activity Bank • Mathematics • 25-35 min

Category	Mathematics	Time	25-35 min
Difficulty	Easy	ID	correlation-illusion-cards

Big idea: Patterns can feel meaningful before they are mathematically justified.

One-Minute Teacher Brief

Students sort paired-variable cards and discover how small samples and salient cases create illusory correlations.

Student Prompt

Inspect Correlation Illusion Cards Stimulus A. Record one first claim, the evidence you used, your confidence, and what would make you revise.

Concrete Stimulus Packet

Correlation Illusion Cards Stimulus A	Student-facing stimulus 1: Cards pair 'used flashcards / did not use flashcards' with 'high quiz score / lower quiz score'. Work the example, then decide whether it gives a pattern, a model, a calculation, or a proof.	Students inspect this exact card first and commit to a claim before hearing anyone else.
Correlation Illusion Cards Stimulus B	Student-facing stimulus 2: A few vivid cards show flashcards plus high scores, while the full set is mixed. Work the example, then decide whether it gives a pattern, a model, a calculation, or a proof.	Use this for comparison, a second round, or a partner challenge.
Correlation Illusion Cards Reveal / Twist	Student-facing stimulus 3: A few vivid cards show flashcards plus high scores, while the full set is mixed. Work the example, then decide whether it gives a pattern, a model, a calculation, or a proof.	Teacher reveal: connect the changed response to the big idea — Patterns can feel meaningful before they are mathematically justified.

Actual Lesson Questions

#	Question for students	Teacher key / cue
1	After inspecting Correlation Illusion Cards Stimulus A, what is your first knowledge claim?	Teacher cue: look for a clear claim, not just a description.
2	Which exact detail from Correlation Illusion Cards Stimulus A did you use as evidence?	Teacher cue: students should name visible words, data, source features, method features, or contextual clues.
3	What did you infer beyond the evidence?	Teacher cue: this is where assumptions, framing, models, or perspective usually appear.
4	How confident are you: 50%, 60%, 70%, 80%, 90%, or 100%?	Teacher cue: confidence is data for the debrief, not a grade.
5	What missing information would most improve the knowledge claim?	Teacher cue: push for method, source, context, comparison, or definitions.
6	When does a perceived pattern become evidence?	Teacher cue: students should move from the classroom example to a general TOK issue.

Student Task

Use the stimulus cards to complete the observation/inference/evidence/confidence grid, then answer one TOK knowledge question connected to Mathematics.

Teacher Key / Reveal

The key move is not the “right answer”; it is the moment students see how patterns can feel meaningful before they are mathematically justified.

- Strong response: separates observation from inference.
- Strong response: names a method, source, model, language choice, context, or perspective.
- Strong response: revises confidence when new evidence or a new criterion appears.
- Weak response: gives only opinion or says “it depends” without criteria.

Student Instructions

- Work silently for the first response so you can notice your own starting point.
- Record one knowledge claim, the evidence behind it, and your confidence level.
- After the reveal or comparison, update your claim in a different colour or column.
- Use the debrief to answer: When does a perceived pattern become evidence?

Setup Checklist

- Build card sets where confirming cases are memorable but not actually dominant.
- Ask for a gut judgement before students count the full table.

Example Stimuli or Materials

- Cards pair 'used flashcards / did not use flashcards' with 'high quiz score / lower quiz score'.
- A few vivid cards show flashcards plus high scores, while the full set is mixed.

Resources To Use

- Resource pack: <resources/correlation-illusion-cards/index.html>
- Card deck CSV: <resources/correlation-illusion-cards/cards.csv>
- Visual prompt: <resources/correlation-illusion-cards/visual-prompt.svg>
- Four-cell table mat for sorting all pairings.
- Salience marker: students star the cards they remembered before counting.

Run Sequence

1. Show a small set of memorable paired examples and ask students to guess the relationship.
2. Give groups the full card set to sort into a four-cell table.
3. Students count all cells, not just confirming cases.
4. Groups decide whether the perceived relationship survives counting.
5. Debrief why humans notice some pairings more than others.

Debrief Moves

- Knowledge question: When does a perceived pattern become reliable mathematical evidence?
- Why are confirming cases often more memorable than disconfirming cases?

- How should knowers protect themselves from pattern hunger?

Teacher Quick Script

- A memorable case is not automatically representative evidence.
- Which cell of the table did your intuition ignore?

Exit Ticket

- One thing I first treated as evidence was...
- My confidence changed because...
- A better knowledge question I can now ask is...